Lecture 9

Covariance and Correlation

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LPO.8800: Statistical Methods for Education Research



Last time



Last time: Tests for comparing two groups

- Confidence interval and test for difference in two population means (μ_1 and μ_2)
- Confidence interval and test for difference in two population proportions (π_1 and π_2)
- Independent vs. dependent samples
- Paired sample t-test
- Statistical power for two-sample tests



Introduction



Introduction

Most everything we have done thus far relates to *univariate* distributions:

- Descriptive statistics (*e.g.*, $\bar{x}, s^2$, centiles)
- Transformations (e.g., z-scores, log)
- Probability distributions (e.g., normal, binomial, uniform) and population parameters (*e.g.*, μ, σ^2)
- Random sampling and inferences about population parameters (confidence intervals, hypothesis testing)



Introduction

The most interesting applications of statistics examine the relationship between two or more variables:

- Do women earn less than men?
- Does education increase earnings?
- Does smoking cause lung cancer?
- Do students learn more in small classes than in large ones?
- Does lower birth weight increase the risk of poor health outcomes later in life?



These are examples of **bivariate analyses**. In Lecture 8 we saw one example of a bivariate analysis, where one variable (the group identifier) was binary. The second variable was either continuous or binary.

Today we will turn to *bivariate* distributions and measures of association, or covariance.



Review of univariate probability distributions



Review of univariate probability distributions

A probability distribution for X assigns probabilities to values of X or to intervals of values. PDFs come in discrete and continuous types, depending on the nature of the random variable.

The PDF $f(x_i)$ for a discrete random variable X provides the probability that $X = x_i$ for possible values of X:

$$f(x) = P(X = x_i)$$

The PDF $f(x)$ for a continuous random variable X provides the probability that X is between certain values. E.g., between a and b:

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$



Review of univariate probability distributions

The probabilities in a discrete PDF are nonnegative and must sum to one. If there are N possible outcomes indexed by i:

$$\sum_{i=1}^{N} P(X = x_i) = 1$$

The area under a continuous PDF (i.e., the integral) must equal one:

$$P(-\infty \leq X \leq +\infty) = \int_{-\infty}^{+\infty} f(x)dx = 1$$



Review of univariate probability distributions

Two important features of a PDF are its **expected value** (mean) and **variance**, the first and second population *moments*.

For a discrete random variable X with N unique outcomes indexed by i, the expected value of X is:

$$E(X) = \mu_X = \sum_{i=1}^{N} x_i P(x_i)$$

The variance of X is:

$$Var(X) = \sigma_X^2 = E(x - E(X))^2 = \sum_{i=1}^{N} (x_i - E(X))^2 P(x_i)$$



Review of univariate probability distributions

For a continuous random variable X, the expected value of X is:

$$E(X) = \mu_X = \int_{-\infty}^{+\infty} x f(x) dx$$

The variance of X is:

$$Var(X) = \sigma_X^2 = E(x - E(X))^2 = \int_{-\infty}^{+\infty} (x - E(X))^2 f(x) dx$$



Joint probability distributions and covariance



Joint probability distributions

A **joint probability distribution** for two random variables X and Y assigns probabilities to values of (X, Y) or to intervals of values. Like univariate PDFs, joint PDFs are discrete or continuous.

The PDF $f(x, y)$ for a pair of discrete random variables X and Y provides the probability that $X = x$ and $Y = y$ for values x and y:

$$f(x, y) = P(X = x \text{ and } Y = y)$$

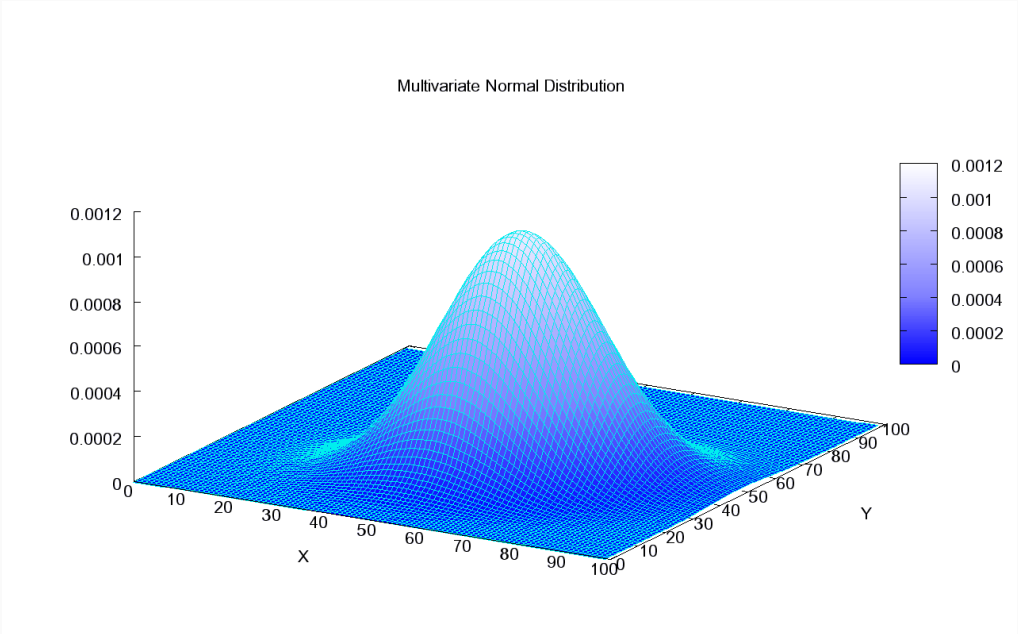
The PDF for continuous random variables X and Y provides the probability that X is between a and b and Y is between c and d:

$$\int_a^b \int_c^d f(x, y) dy dx$$



Example: bivariate normal distribution

An example joint PDF: bivariate normal distribution



Joint probability distributions

The probabilities in a discrete joint PDF are nonnegative and must sum to one. If there are N possible outcomes for X indexed by i, and M possible outcomes for Y indexed by j:

$$\sum_{i=1}^{N}\sum_{j=1}^{M} P(X = x_i \text{ and } Y = y_j) = 1$$

The area under a continuous joint PDF (i.e., the double integral) must equal one:

$$P(-\infty \leq X \leq +\infty \text{ and } -\infty \leq Y \leq +\infty) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y)dydx = 1$$

Measures of association

Bivariate probability distributions allow us to think about how two variables are related in the population:

- *Direction*: are the variables positively correlated, negatively correlated, or uncorrelated?
- *Shape*: is the relationship between the two variables *linear* or *nonlinear*?
- *Strength*: is the relationship between the two variables strong, weak, or moderate?

The joint PDF tells us nothing about whether the relationship is a *causal* one or not.



Covariance

The population **covariance** between two random variables X and Y is:

$$Cov(X, Y) = \sigma_{xy} = E\left[(x - E(X))(y - E(Y))\right]$$

For a discrete joint PDF:

$$Cov(X, Y) = \sigma_{xy} = \sum_{x} \sum_{y} (x - E(X))(y - E(Y))f(x, y)$$

For a continuous joint PDF:

$$Cov(X, Y) = \sigma_{xy} = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (x - E(X))(y - E(Y))f(x, y)dydx$$



Covariance

The covariance is like a weighted average of products: X's deviation from its mean multiplied by Y's deviation from its mean.

- If Y tends to be higher than average when X is higher than average, these products will tend to be positive (a **positive covariance**).
- If Y tends to be *lower* than average when X is higher than average, these products will tend to be negative (a **negative covariance**).

The weights come from the joint probability of X and Y.



Covariance

Some facts about covariance:

- Magnitude of the covariance depends on the units of X and Y
- $Cov(X, Y) = E(XY) - E(X)E(Y)$
- If X and Y are independent, $Cov(X, Y) = 0$
- Independence implies zero covariance but the converse is not true.
- Covariance is a measure of *linear* association
- If $Y = X$, $Cov(X, X) = Var(X)$. (The covariance of a variable with itself is just its variance).



Correlation

Correlation is a standardized, or unit-free measure of covariance:

$$Corr(X, Y) = \rho_{xy} = E\left[\left(\frac{x - E(X)}{\sigma_x}\right)\left(\frac{y - E(Y)}{\sigma_y}\right)\right]$$

$$= \frac{Cov(X, Y)}{\sigma_x \sigma_y} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$



Correlation

Some facts about correlation:

- Like covariance, ρ is a measure of *linear* association
- $-1 \leq \rho_{xy} \leq 1$
  - $\rho_{xy} = 1$ is a perfect positive correlation
  - $\rho_{xy} = -1$ is a perfect negative correlation
  - $\rho_{xy} = 0$ is no correlations
- ρ_{xy} requires both σ_x and σ_y to be positive (i.e., not zero).

Later: estimation of population ρ using a sample.



Scatter diagrams



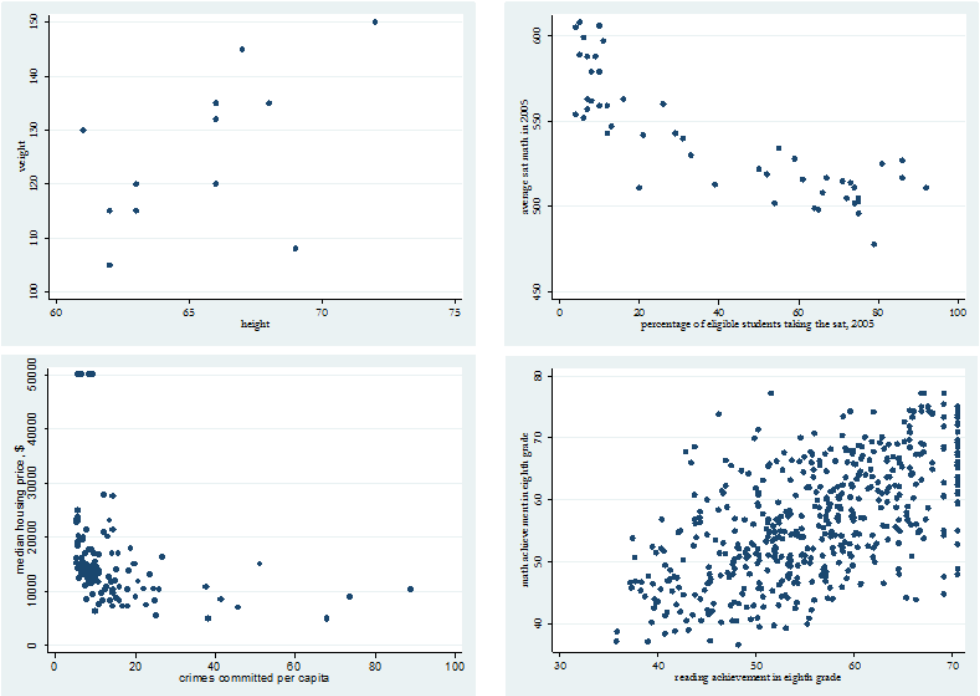
Scatter diagrams

The easiest way to see how two variables are associated using data is via a *scatter diagram* or *scatterplot*.

- In Stata: `scatter` *`yvar xvar`*
- Appropriate for variables that are at least interval measured
- Scatter diagrams can provide a sense of *direction* of relationship (if any), *linearity*, and *strength* of association



Scatter diagrams



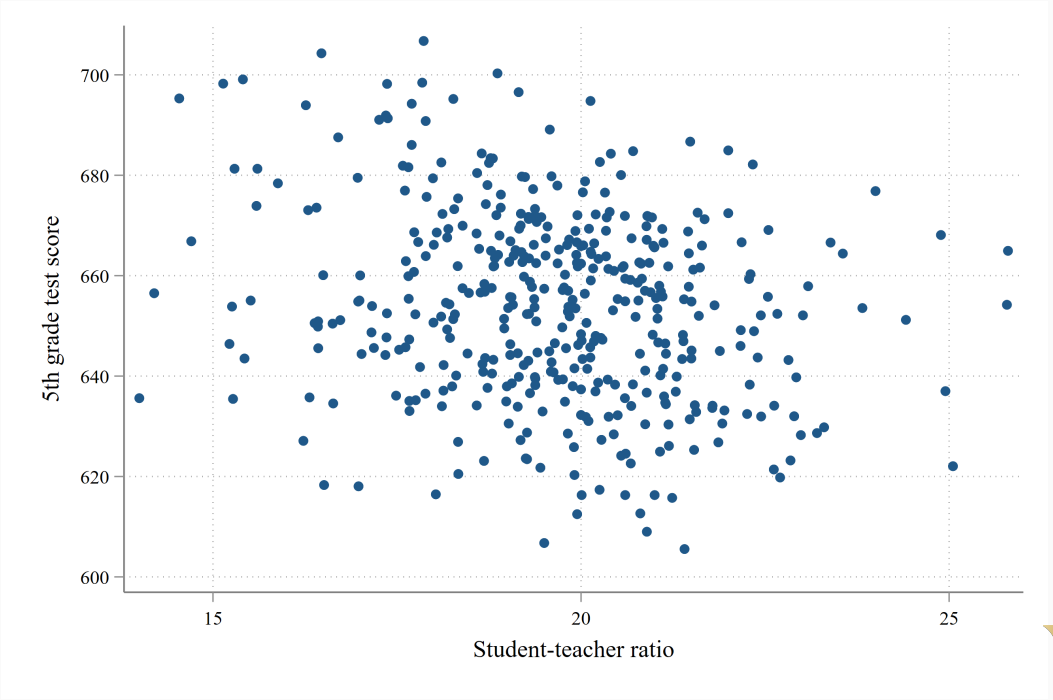
Scatter diagrams

Often with scatter diagrams there is a natural **response** or **outcome**, and **explanatory** variable. We may have in mind a theory in which variation in the response is (at least in part) explained by variation in the explanatory variable.

- Denote the outcome as *Y* and explanatory variable as *X*.
- Alternatively, "dependent" and "independent" variables. (I prefer *not* to use these, given these terms' other meanings in statistics).

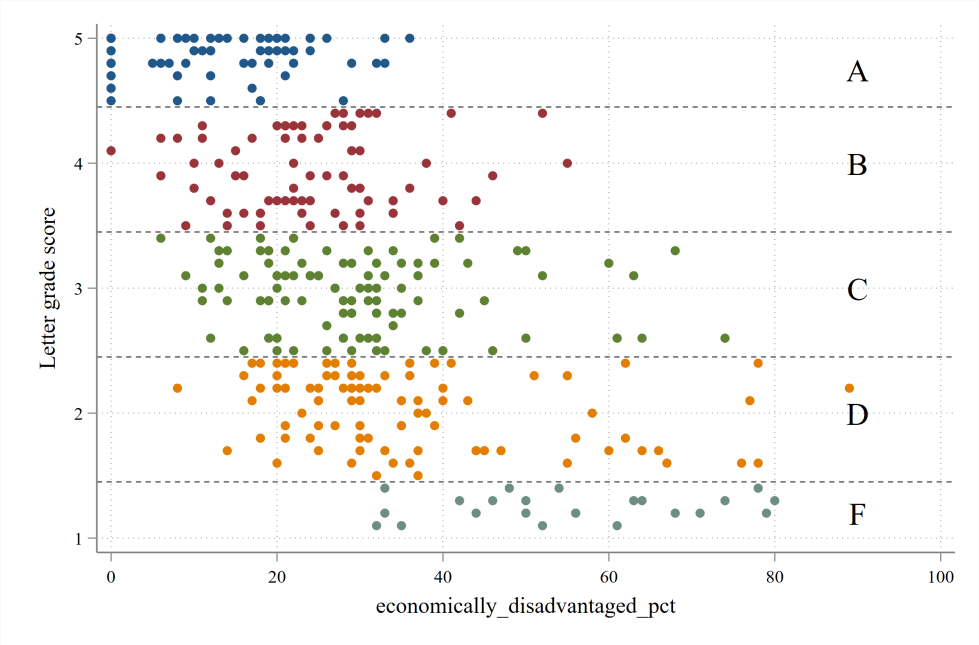
Scatter diagram 1: California test scores and class size

From the *caschool* data (California school districts in 1998-99): 5th grade test scores vs. student-teacher ratio



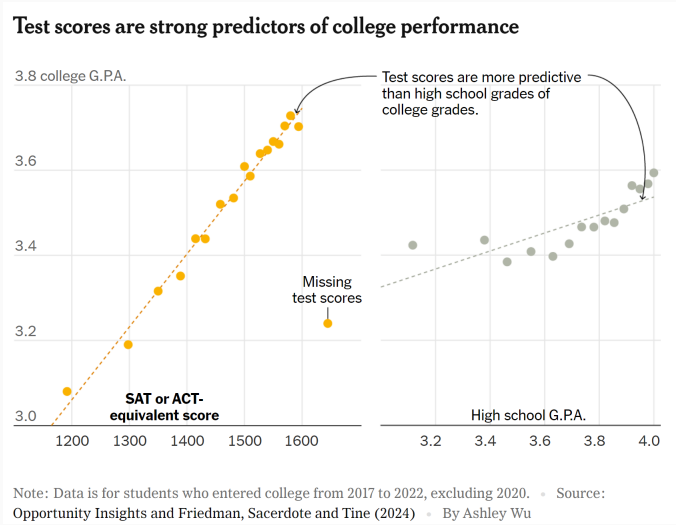
Scatter diagram 2: school letter grades in Tennessee

From the *TN-lettergrades* data: overall scores vs. percent economically disadvantaged, Tennessee high schools in 2022-23.



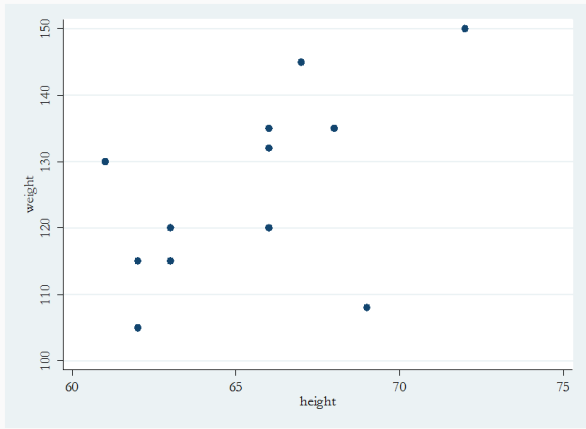
Scatter diagram 3: predictors of college performance

Predictors of college performance (SAT/ACT vs. high school GPA)



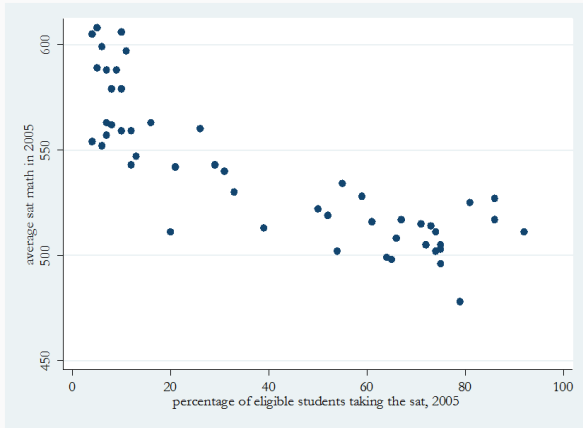
Source: David Leonhardt, "The Misguided War on the SAT," *The New York Times*, January 7, 2024.
Note: these are binned, not raw, data. The data come from highly selective universities.

Scatter diagram 4: height and weight



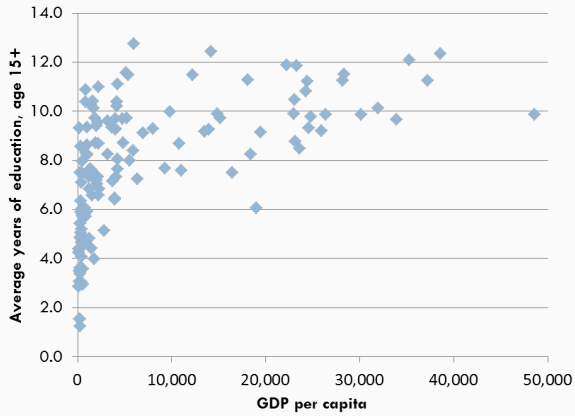
- Positive association
- Mostly linear–a line would describe this relationship well
- Moderately strong association

Scatter diagram 5: percent taking SAT and scores



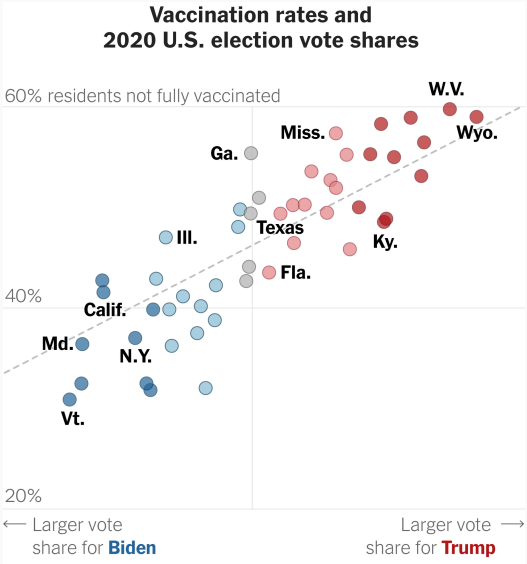
- Negative association
- Mostly linear (some curvature)
- Strong association

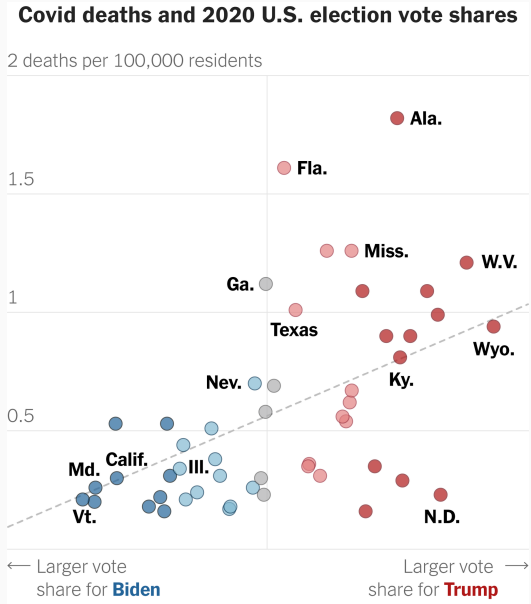
Scatter diagram 6: GDP per capita and education



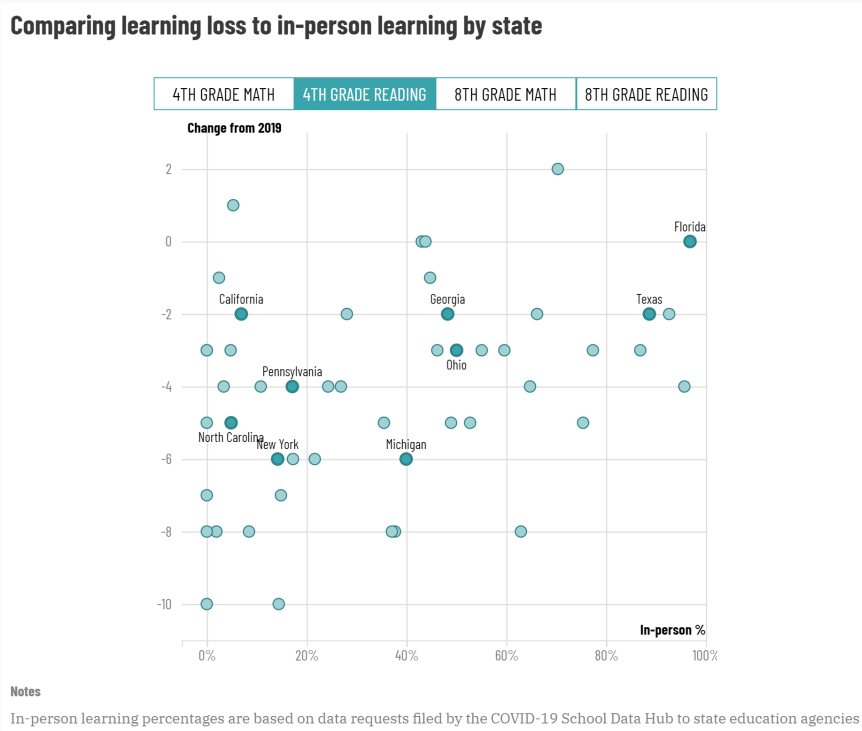
- Positive association
- *Nonlinear* relationship
- Moderately strong association

Scatter diagram 7: Politics and COVID

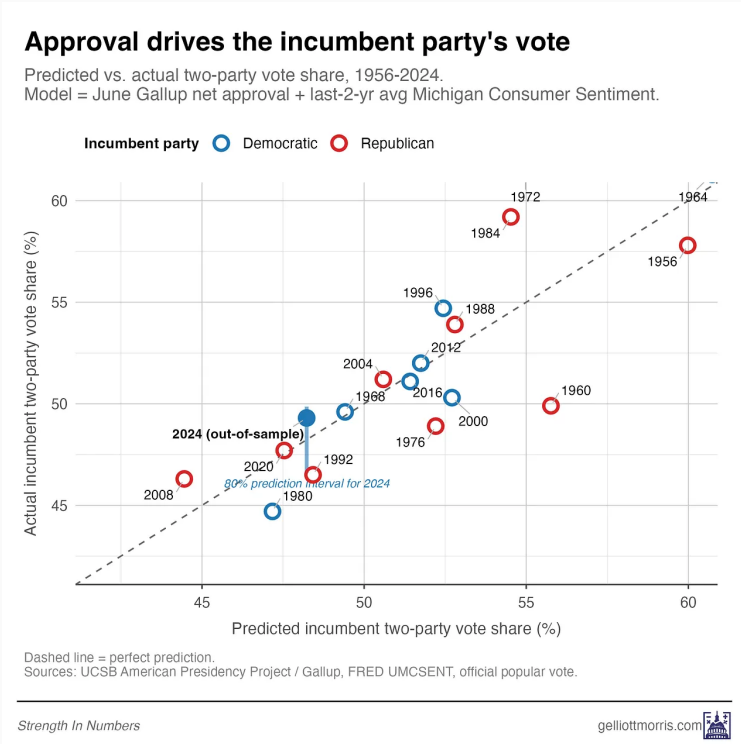




Scatter diagram 8: In-person learning

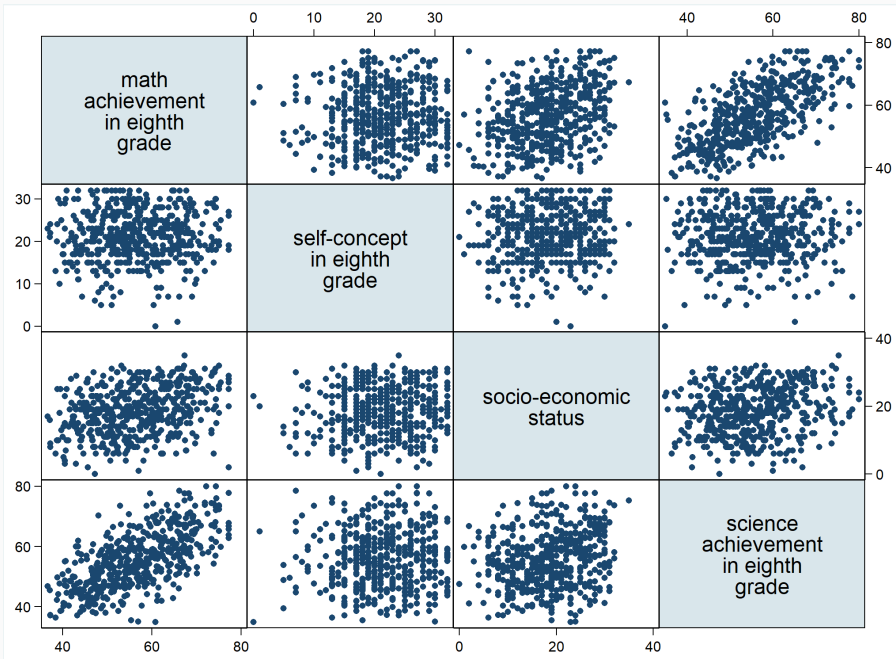


Scatter diagram 9: Incumbent voting



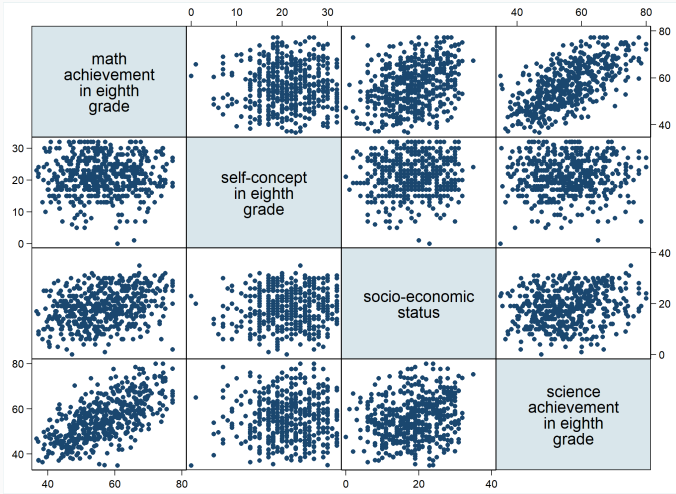
Scatterplot matrix

`graph matrix` *`varlist`* is a useful command for visualizing the bivariate associations between two or more variables. For example:



Scatterplot matrix

The horizontal axes in each column apply to the variable named in that column; the vertical axes apply to the variable named in that row. Thus, for example, the scatterplot in cell (3,1) is the same as the scatterplot in cell (1,3), but with the axes flipped.



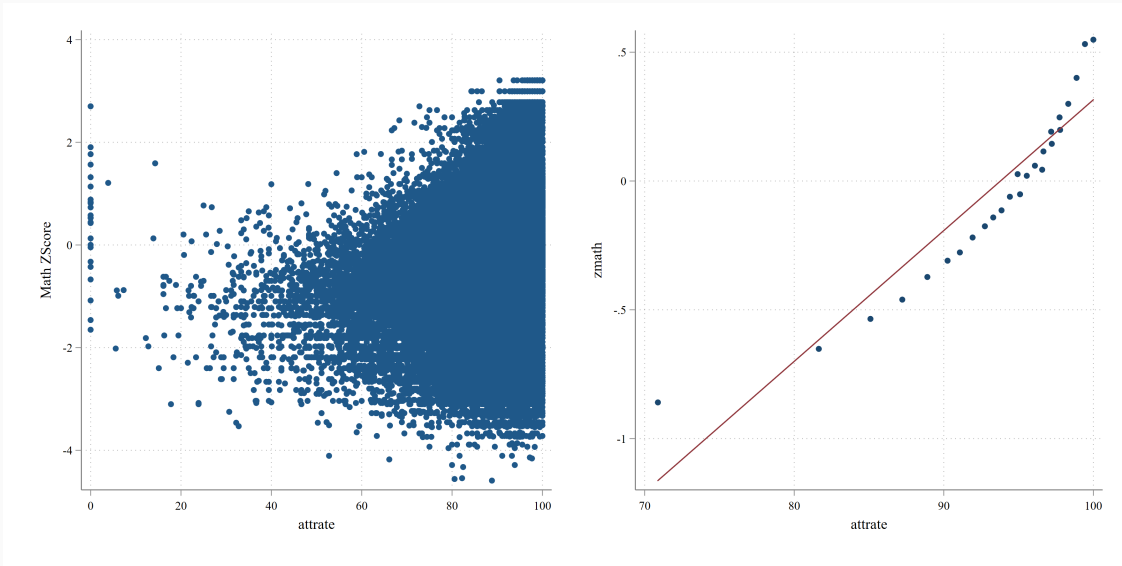
Binscatter

With large datasets scatterplots may be difficult to read. The user-written command **`binscatter`** groups the x-axis into bins and computes the mean for each bin. The command has lots of options, including a fitted line.

```
binscatter yvar xvar, options
```

You can change the number of bins (default is 20), whether the fitted line is linear or quadratic or connected, etc.

Math scores and annual school attendance in a large dataset ($N > 420,000$)



Sample covariance and correlation

Sample covariance and correlation

The sample analog to the population covariance is:

$$s_{xy} = \frac{\sum_{i=1}^{N} (x_i - \bar{x})\,(y_i - \bar{y})}{n - 1}$$

The sample analog to the population *correlation* is:

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{s_x s_y (n-1)} = \frac{s_{xy}}{s_x s_y}$$

r_{xy} is known as the sample *correlation coefficient* or the Pearson product moment correlation. The sample s_{xy} and r_{xy} are estimators of the population parameters σ_{xy} and ρ_{xy}.



Covariance and correlation—calculation

	Height (x_i)	Weight (y_i)	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x}) \times (y_i - \bar{y})$
	69	108	3.58	-17.83	-63.90
	61	130	-4.42	4.17	-18.40
	68	135	2.58	9.17	23.68
	66	135	0.58	9.17	5.35
	66	120	0.58	-5.83	-3.40
	63	115	-2.42	-10.83	26.18
	72	150	6.58	24.17	159.10
	62	105	-3.42	-20.83	71.18
	62	115	-3.42	-10.83	37.01
	67	145	1.58	19.17	30.35
	66	132	0.58	6.17	3.60
	63	120	-2.42	-5.83	14.10
Mean	65.42	125.83	0.00	0.00	23.74
Sum			0.00	0.00	284.85
SD	3.32	14.24			



Covariance and correlation—calculation

$$s_{xy} = \frac{\sum_{i=1}^{N} (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$s_{xy} = \frac{284.85}{12-1} = 25.895$$

$$r_{xy} = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{s_x s_y (n-1)}$$

$$r_{xy} = \frac{284.85}{(3.32)(14.24)(12-1)} = 0.548$$



Covariance and correlation in Stata

To obtain correlation coefficients in Stata, use `corr` *`yvar xvar1 xvar2...`*. The result is a correlation matrix.

- Be aware of how Stata handles missing values:
  - **listwise deletion** means observations are not used if *any* of the listed variables in the command are missing.
  - **pairwise deletion** means correlations of pairs of variables are considered in isolation.
- `pwcorr` *`yvar xvar1 xvar2...`* uses pairwise deletion. `corr` uses listwise deletion.

To obtain the covariance in Stata, use `corr` with `cov` option. The result is a **variance-covariance matrix**.



Correlation coefficient in Stata

Examples:

```
. corr height weight
(obs=12)
```

	height	weight
height	1.0000	
weight	0.5486	1.0000

```
. corr height weight, cov
(obs=12)
```

	height	weight
height	10.9924	
weight	25.8939	202.697



Correlation coefficient in Stata

Examples:

```
. corr achmat08 achrdg08
(obs=500)
```

	achmat08	achrdg08
achmat08	1.0000	
achrdg08	0.5947	1.0000

```
. pwcorr achmat08 achrdg08 achmat10 achrdg10
```

	achmat08	achrdg08	achmat10	achrdg10
achmat08	1.0000			
achrdg08	0.5947	1.0000		
achmat10	0.8489	0.5919	1.0000	
achrdg10	0.5803	0.7538	0.6531	1.0000



Correlation coefficient

What is a "strong correlation?" It depends on the context. (How strong would you expect the correlation to be? Is there a theoretical reason why the correlation should be particularly strong or weak?)

- Rule of thumb ("Cohen's scale") based on $|r_{xy}|$:
  - $|r_{xy}| < 0.1$: zero to weak correlation
  - $0.1 < |r_{xy}| < 0.3$: weak to moderate correlation
  - $0.3 < |r_{xy}| < 0.5$: moderately strong correlation
  - $|r_{xy}| > 0.5$: strong correlation

The correlation coefficient itself is *ordinal*. An increase in correlation from 0.1 to 0.2 is not equivalent to an increase from 0.4 to 0.5.

Try "guess the correlation:"

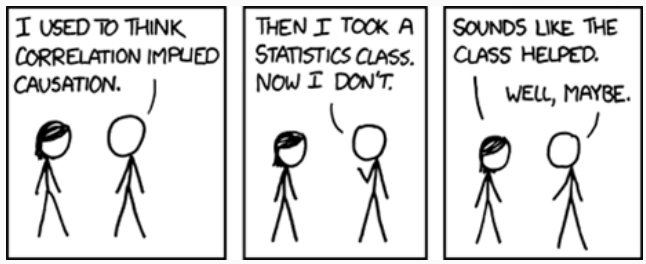
`https://istats.shinyapps.io/guesscorr/`



Correlation vs. causation

Important: *correlation does not imply causation!*

- *Correlation* means two variables move together.
- *Causation* means that change in one variable is causing change in the other.

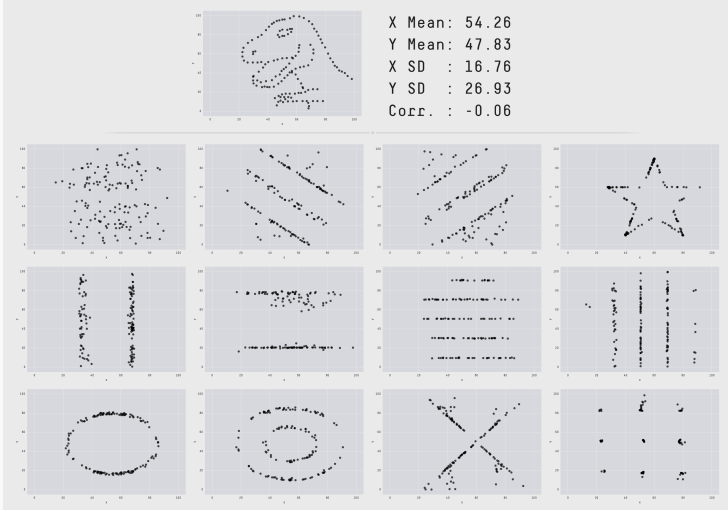






The importance of visualizing your data

Never trust summary statistics alone! All of the datasets used below have the same $\bar{x}, \bar{y}, s_x, s_y$, and r_{xy}.



Recommended source:

`https://www.autodesk.com/research/publications/same-stats-different-graphs`

Correlation: special cases

"Point biserial" correlation: when one variable (x) is dichotomous and the other (y) is continuous:

$$r_{xy} = \frac{(\bar{y}_1 - \bar{y}_0)s_x}{s_y}$$

$\bar{y}_1$ is the mean of y for observations where $x = 1$, and $\bar{y}_0$ is the mean of y for observations where $x = 0$.

"Phi coefficient": when both variables are dichotomous, r can be calculated using counts from a 2 x 2 table:

$$r_{xy} = \phi = \frac{AD - BC}{\sqrt{(A+B)(C+D)(A+C)(B+D)}}$$

	y	
x	0	1
0	A	B
1	C	D



Spearman rank correlation



Spearman rank correlation

Spearman's rank correlation (sometimes called "rho") can be used with ordinal-measured variables—where the size of the difference between x and its mean is not meaningful—or in cases where the underlying relationship is nonlinear.



Spearman rank correlation

Spearman's correlation depends on how variables *rank* in their respective distributions.

- Rank each variable x and y in its respective distribution, in ascending order $1, \ldots, n$
- For an observation i, d_i is the *difference* between i's ranking for x and i's ranking for y: $d_i = rank(x_i) - rank(y_i)$

$$r_{s,xy} = 1 - \frac{6 \sum_{i=1}^{n} d_i^2}{n^3 - n}$$



Spearman rank correlation

A few notes about Spearman's correlation:

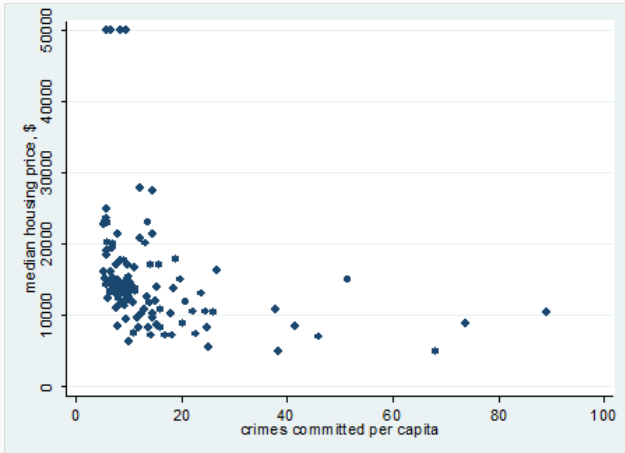
- When x_i and y_i are identically ranked, $d_i = 0$. If $d_i = 0$ for *all cases*, then $r_{s,xy} = 1$ (a perfect positive correlation).
- The further apart x_i and y_i are in their ranks, the larger is d_i and $r_{s,xy}$ gets closer to -1 (a perfect negative correlation).

$$r_{s,xy} = 1 - \frac{6 \sum_{i=1}^{n} d_i^2}{n^3 - n}$$



Spearman rank correlation in Stata

The Spearman rank correlation is obtained in Stata with the command `spearman` *`yvar xvar`*. For example, consider the nonlinear relationship between median house price and crime:





Spearman rank correlation in Stata

```
. spearman price crime

 Number of obs =      506
Spearman's rho =   -0.5564

Test of Ho: price and crime are independent
    Prob > |t| =    0.0000

. sum price crime

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
       price |        506    22511.51    9208.856       5000      50001
       crime |        506    3.611536    8.590247       .006     88.976

. corr price crime
(obs=506)

             |    price    crime
-------------+------------------
       price |   1.0000
       crime |  -0.3879   1.0000
```



Variable transformations and correlation



Variable transformations and correlation

Consider the linear transformations of variables x and y:

$$x_1 = a + (b * x_0)$$
$$y_1 = c + (d * y_0)$$

What happens to the correlation between two variables x and y when one (or both) is transformed by a linear function?



Variable transformations and correlation

Re-write the correlation coefficient as follows.

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{s_x s_y (n-1)} = \frac{\sum_{i=1}^{n}(\frac{x_i - \bar{x}}{s_x})(\frac{y_i - \bar{y}}{s_y})}{n-1} = \frac{\sum_{i=1}^{n} z_{xi} z_{yi}}{n-1}$$

How is a z-score affected by a linear transformation?

$$z_{x_1} = \frac{a + (b * x_0) - a - (b * \bar{x_0})}{|b| * s_{x_0}} = \frac{b * (x_0 - \bar{x_0})}{|b| * s_{x_0}} = \frac{b}{|b|} z_{x_0}$$

When b is positive, the z-score of the transformed variable x_1 is the same as the original z-score. When b is *negative*, the new z-score is the inverse of the original. Note a and c cancel out.



Variable transformations and correlation

How transformations affect the correlation between x and y thus depend only on the multiplicative factor applied to x, y, or both:

- If both multiplicative factors are positive ($b > 0$ and $d > 0$) or both multiplicative factors are negative ($b < 0$ and $d < 0$), then the correlation between the new, transformed variables is the same as the correlation between the original variables.
- If one multiplicative factor is positive and the other is negative ($b > 0$ and $d < 0$ or ($b < 0$ and $d > 0$), then the correlation between the new, transformed variables is the negative of the correlation between the original variables.



Variable transformations and covariance

Plug the transformed values of x_0 and y_0 into the covariance:

$$
\begin{aligned}
s_{xy} &= \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n-1} \\
s_{xy,new} &= \frac{\sum_{i=1}^{n}(a + bx_{0i} - a - b\bar{x}_0)(c + dy_{0i} - c - d\bar{y}_0)}{n-1} \\
&= \frac{\sum_{i=1}^{n}(bx_{0i} - b\bar{x}_0)(dy_{0i} - d\bar{y}_0)}{n-1} \\
&= \frac{\sum_{i=1}^{n} bd(x_{0i} - \bar{x}_0)(y_{0i} - \bar{y}_0)}{n-1} \\
&= bds_{xy,old}
\end{aligned}
$$

That is, the covariance of the two transformed variables is equal to the covariance of the original (untransformed) variables times bd.



Significance testing for correlation



Hypothesis tests about ρ

The sample correlation r_{xy} can be used to estimate the population correlation coefficient ρ. If we know the sampling distribution of r_{xy}, we can construct confidence intervals for ρ or conduct hypothesis tests.

Under $H_0 : \rho = 0$, the following test statistic has a t-distribution with $n - 2$ degrees of freedom:

$$t = \frac{|r_{xy}|\sqrt{n-2}}{\sqrt{1 - r_{xy}^2}}$$

Find the p-value for t and then determine whether $p < \alpha$. In Stata, use `pwcorr` with the `sig` option. Note this t statistic is $|r - 0|$ divided by the standard error



Hypothesis tests about ρ

Example:

```
. pwcorr height weight, sig

             |   height   weight
-------------+------------------
      height |   1.0000
             |
             |
      weight |   0.5486   1.0000
             |   0.0648

. display (0.5486*sqrt(10))/sqrt(1-0.5486^2)
2.0749393

. display 2*ttail(10,2.0749393)
.06474521
```

0.0648 is the p-value for the two-sided hypothesis test with $H_0 : \rho = 0$.



Hypothesis tests about ρ

Assumptions:

- Independent random sample from the population
- x and y follow a bivariate normal distribution
- If sample size is large enough, normality is not required and test statistic follows an approximate t distribution.



Hypothesis tests about ρ

Note the `pwcorr` includes a `bonferroni` option that adjusts p-values for multiple hypotheses:

```
. pwcorr price mpg headroom weight gear_ratio, sig bonf
```

	price	mpg	headroom	weight	gear_r~o
price	1.0000				
mpg	-0.4686 0.0003	1.0000			
headroom	0.1145 1.0000	-0.4138 0.0025	1.0000		
weight	0.5386 0.0000	-0.8072 0.0000	0.4835 0.0001	1.0000	
gear_ratio	-0.3137 0.0650	0.6162 0.0000	-0.3779 0.0090	-0.7593 0.0000	1.0000

p-values have been multiplied by the number of pairwise tests $g(g-1)/2$, where g is the number of variables. Above: 10



Hypothesis tests about ρ

Confidence intervals for ρ are not part of the `corr` command. It is not as straightforward as using the t test statistic above to construct the interval, since this is the distribution under the null of no correlation.

When $|\rho| > 0$, the sampling distribution is actually quite *skewed*, even with large samples. This is because the correlation coefficient is constrained to be between -1 and +1.



Fisher's z

Fisher showed that a transformation of r has an approximate normal distribution (the inverse hyperbolic tangent transformation):

$$z' = 0.5 \ln \left[\frac{1+r}{1-r} \right]$$

The standard error of z' is $1/\sqrt{n-3}$. You can calculate a $(1-\alpha)\%$ confidence interval using z' and then transform back into the correlation scale.

This is not a built-in Stata function, but see the user-written commands `corrci` and `corrcii` and Cox (2008).

Cox (2008): `https://journals.sagepub.com/doi/pdf/10.1177/1536867X0800800307`



Simulating draws from a bivariate normal in Stata

Example of 100 draws of X and Y from a bivariate normal distribution with $\mu_x = \mu_y = 0$, $\sigma_x^2 = \sigma_y^2 = 1$ and $\rho_{xy} = 0.5$:

```
matrix C = (1, 0.5 \ 0.5, 1)
drawnorm x y, n(100) corr(C)
```

The `corr` option in `drawnorm` takes a matrix which tells Stata how the two simulated variables are correlated. Note the covariance in this case is also 0.5 since $\rho = \sigma_{xy}/\sigma_x\sigma_y$ and $\sigma_x^2 = \sigma_y^2 = 1$.

$$C = \begin{bmatrix} 1.0 & 0.5 \\ 0.5 & 1.0 \end{bmatrix}$$



Chi-squared test of indpendence



Test for association between categorical variables

The (Pearson) correlation is appropriate for interval-meausured variables[1] and the Spearman rank correlation works for ordinal variables. What about *categorical* variables?

	Party ID			
Gender	Democrat	Independent	Republican	Total
Female	573	516	422	1511
Male	386	475	399	1260
Total	959	991	821	2771

[1]Though we saw that one or both can be binary.



Test for association between categorical variables

We can consider the test:

H_0 : the variables are statistically independent
H_1 : the variables are not independent

If the two variables in the above example are independent, then the distribution of *Party ID* by *gender* should be identical in the population. In a sample, we are unlikely to see exactly identical distributions. How large a difference would lead us to reject H_0?



Chi-squared test of independence

Consider the test statistic below, which uses observed frequencies (O_{ij}) and expected frequencies (E_{ij}) in row i and column j:

$$\chi^2 = \sum_{ij} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

The sum is over all combinations of the two variables. The expected frequencies E_{ij} are those we would expect to see if the share in each marginal category mirrored the combined sample.

When H_0 is true, we expect O_{ij} and E_{ij} to be close and χ^2 to be small. We reject H_0 if χ^2 is "large enough."



Chi-squared distribution

With large samples, the sampling distribution of this test statistic is χ^2 with $(r-1)(c-1)$ degrees of freedom (r =rows and c =columns).

About the χ^2 distribution:

- It only includes non-negative values.
- It is right-skewed.
- Its shape depends on the degrees of freedom (df)
- It has a mean of $\mu = df$ and standard deviation $\sqrt{2df}$. So, as $df \uparrow$ the mean shifts to the right and the disperson increases.

In Stata, to get $Pr(\chi^2 > c)$ use `chi2tail(df,c)`



Chi-squared distribution

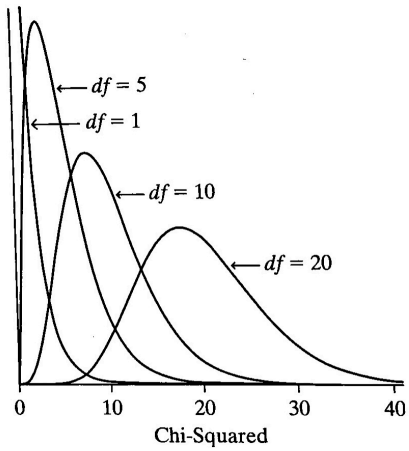


FIGURE 8.2: The Chi-Squared Distribution. The curve has larger mean and standard deviation as the degrees of freedom increase.



Chi-squared test in Stata

In Stata, can get χ^2 and p-value:

```
. tabulate gender edexpect, chi2
```

gender	highest level of education expected: less than	bachelor'	master's	ph.d., md	Total
male	20	75	81	51	227
female	28	84	109	52	273
Total	48	159	190	103	500

Pearson chi2(3) = 1.7617 Pr = 0.623

```
. tabulate algebra8 edexpect, chi2
```

algebra taken in eighth grade	highest level of education expected: less than	bachelor'	master's	ph.d., md	Total
no	34	79	73	42	228
yes	11	65	102	56	234
Total	45	144	175	98	462

Pearson chi2(3) = 19.8478 Pr = 0.000



Chi-squared test in Stata

Using table calculator (`tabi`) for *Party ID* by *gender*:

```
. tabi 573 516 422 \ 386 475 399, chi2
```

row	col: 1	2	3	Total
1	573	516	422	1,511
2	386	475	399	1,260
Total	959	991	821	2,771

Pearson chi2(2) = 16.2017 Pr = 0.000

You can also request row and column percentages, expected frequencies, and more.



More on the Chi-squared test of independence

Assumptions:

- Observations are independent
- Expected cell frequencies are sufficiently large (≥ 5).

Interpretation:

- The test is whether there is an association; it does not measure *strength* of association.



Next time



Next time

Linear regression (single variable predictor):

- "Line of best fit"
- Least squares (OLS)
- Regression for prediction
- Predicted values and residuals
- Measuring fit (R^2)

